

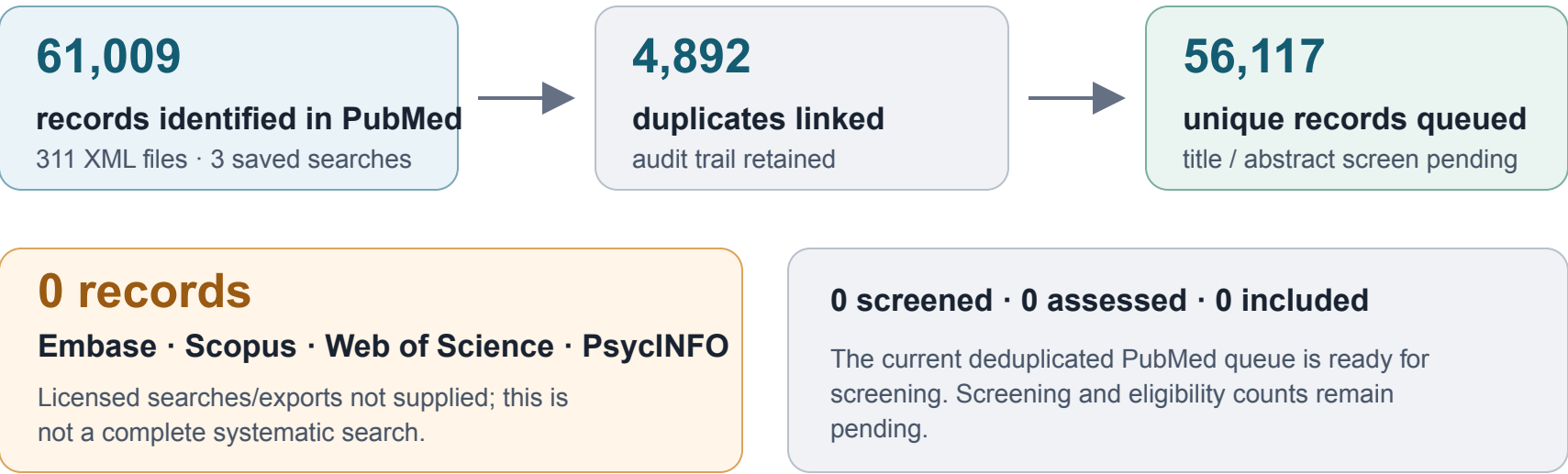
Study framework and current evidence status

Sleep and circadian phenotypes × frailty definitions × aging context

WORKING FIGURE · REVIEW SCREENING NOT STARTED

A Evidence acquisition

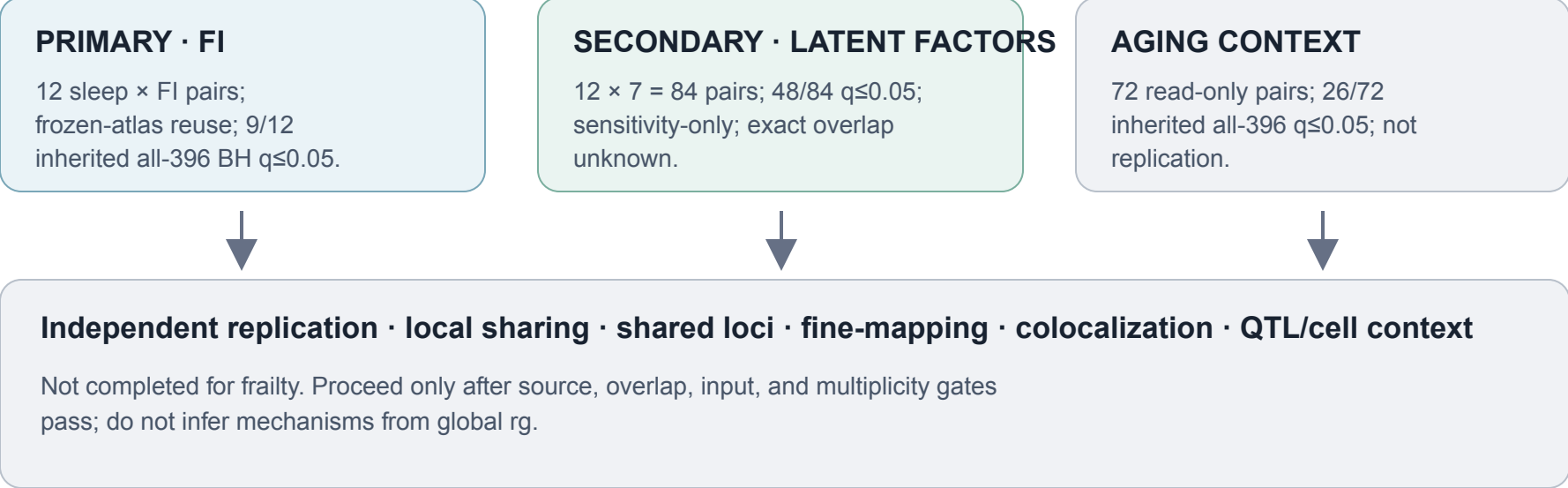
PubMed-only snapshot; manual database exports are still absent



Counts are a reconciliation of retained XML and deduplicated records, not completed screening outcomes.

B Study design and analysis sequence

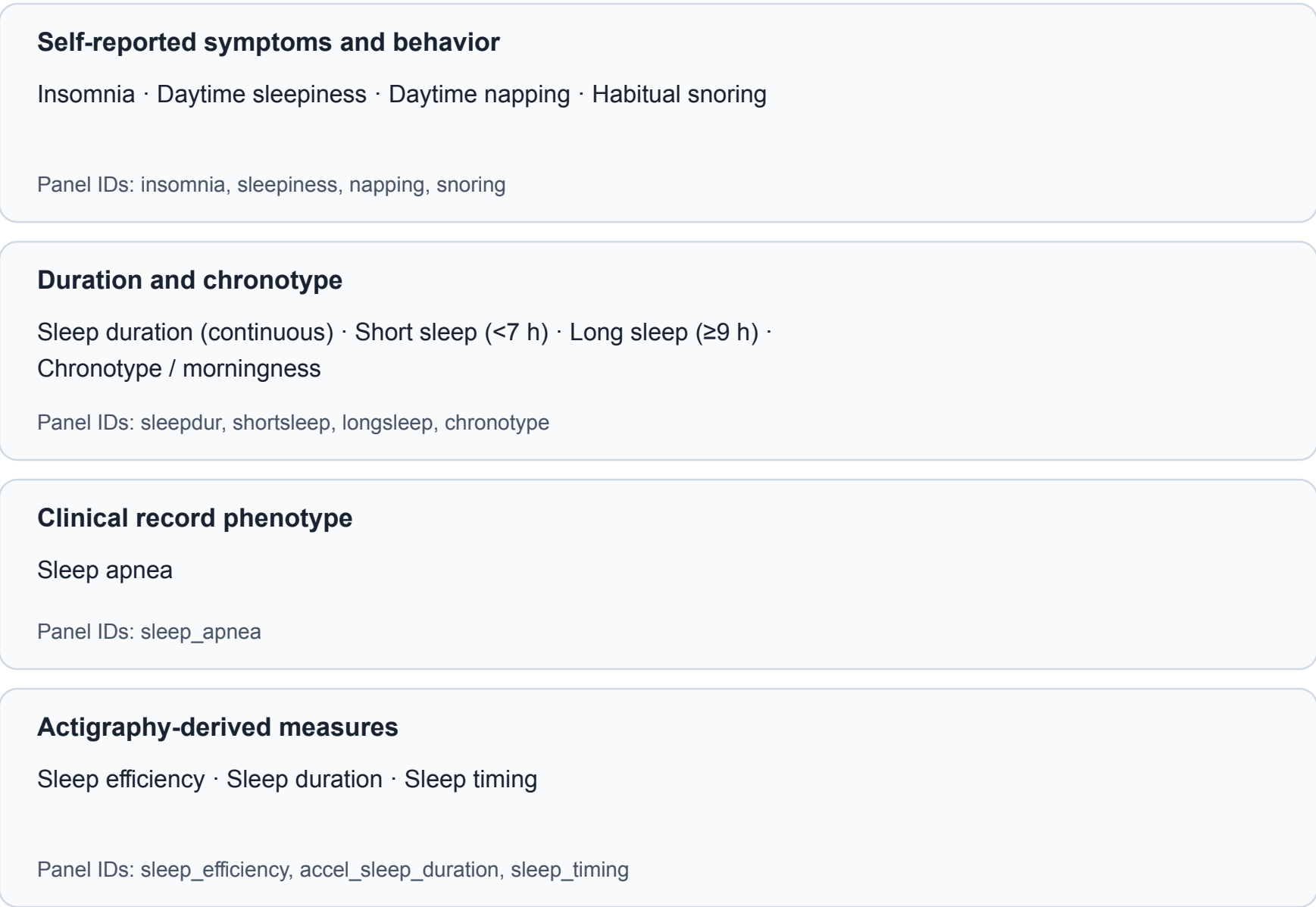
Distinct families are kept separate; correlations are not causal effects



UK Biobank overlap is expected in 11/12 primary pairs and in latent-family sources; exact intersections remain unknown.

C Locked sleep and circadian panel

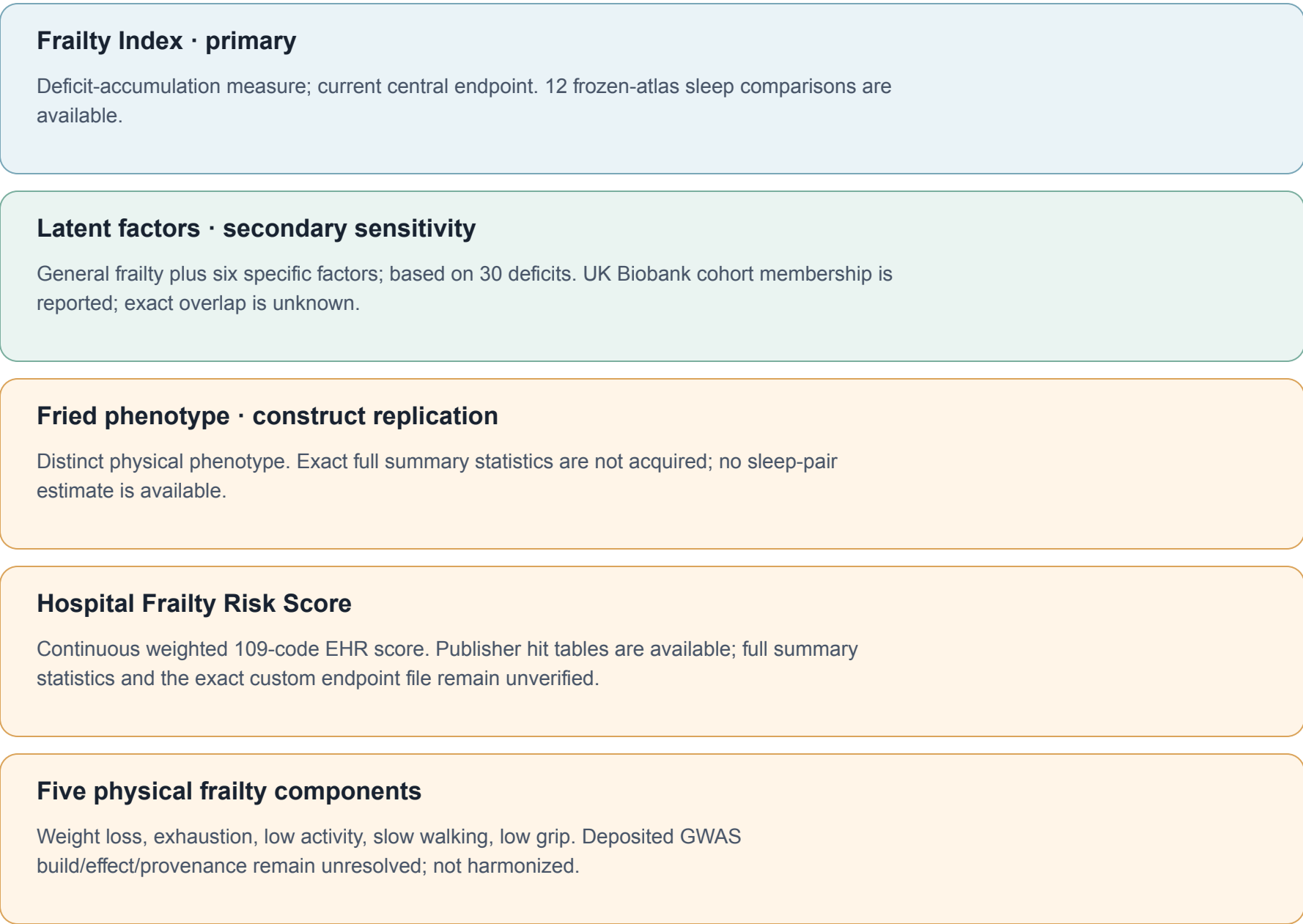
12 registered phenotypes; groups organize measures without treating them as interchangeable



Panel taxonomy follows the locked atlas-v1.0 source registry.

D Frailty definitions and evidence status

The constructs overlap but are not interchangeable



No eligible physical-component, Fried, or HFRS sleep analysis is represented here.

INTERPRETATION Genome-wide genetic correlation describes shared effects; it does not establish causation, independent replication, local sharing, or mechanism. Figure 1 remains provisional until licensed searches, dual screening, eligibility assessment, and the reporting flow are complete.